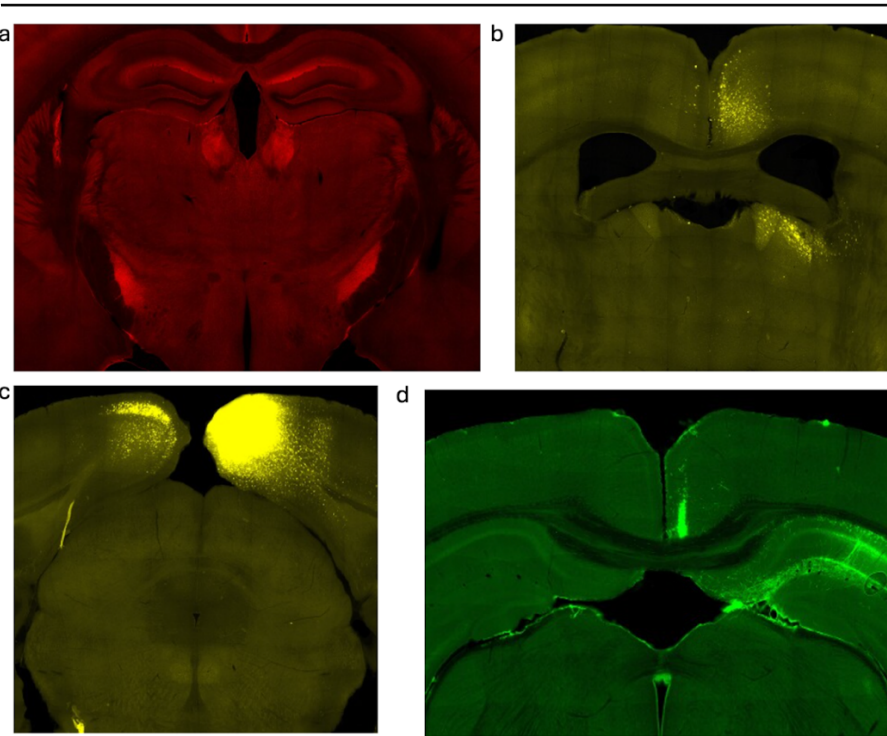


Retrosplenial Dendritic Atrophy Drives Dissociation by Impairing Subthreshold Integration



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Retrosplenial atrophy drives dissociation, yet the mechanism is unknown. I propose a longitudinal, within-subject, all-optical design linking dendritic atrophy to subthreshold integration failure. Using shRNA-induced atrophy and voltage imaging, I aim to prove that structural degradation uncouples sensation from action, establishing a causal target for potential therapeutic rescue.

Figure 1: The Anterograde and Retrograde Traces of RSC. a, AAV-RFP confirms dense RSPd projections to Anterior Thalamus (AV). b, FG reveals the thalamic return loop. c, FG highlights Visual Cortex (V1) inputs to the RSC. d, Anterograde tracing reveals Subiculum (SUB) inputs to the Retrosplenial Cortex (RSC). (Mouse Connectome Project).

Methods: Triple-AAV RSC coinjection (CreER-LSL-shKal7, ASAP5, codon-optimized rescue) with retrograde V1 and anterograde AV. Drifting-grating contextual go/no-go task (A: Red/Odor1/Click; B: Blue/Odor2/Silence) with 10% conflict probes. 2P imaging (920/1040)

track subthreshold dynamics (dF/F_0 , PSD) across atrophy stages. Isoflurane for structural stack after each recording. Retroorbital injection and fiducial burns will be done before death. Brains to be cleared with SHIELD and registered via vascular landmarks and reverse-warping.

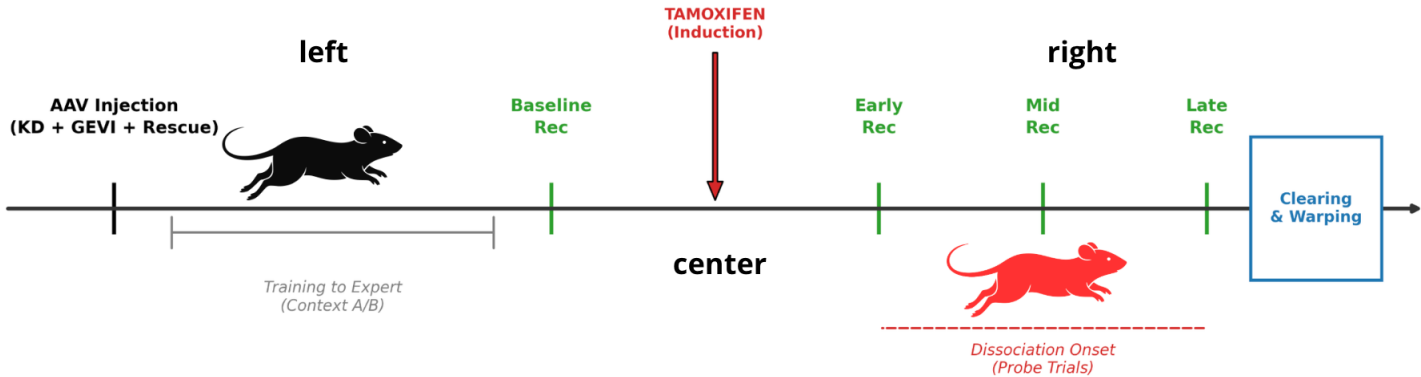


Figure 2: Experimental Timeline. **left**, mice are trained to expert mice. **center**, Tamoxifen induces KD and Rescue. **right**, Dissociation is hypothesized to emerge post-induction, characterized by declining accuracy and pupillary dilation on probe trials. Tissue clearing ensures deficits are driven by dendritic atrophy, not upstream or downstream projection pathology.